

WaterCam PCB HAT - Connection Map

This document shows every electrical connection on the WaterCam custom PCB, written so you can see at a glance what's connected to what.

Reading the arrows: A - -> B means A is electrically connected to B. If multiple things are tied together, they appear on the right of the arrow separated by commas.

Components on the board:

Name	What it is
Pi	Raspberry Pi 4B GPIO header (40 pins, mates with the WittyPi underneath)
WittyPi	WittyPi 4 power management module (sits between Pi and our hat)
AHT20	Adafruit temperature/humidity sensor breakout
BNO055	Adafruit 9-axis IMU breakout
FLIR Lepton	Thermal camera socket
MTDOT	MultiTech LoRa radio module
MOSFET	IRLB8721PBF NMOS transistor (acts as a switch for the wake signal)
resistor	10k Ω resistor (keeps the MOSFET turned off when no signal is present)

Power and Ground

5V power rail

- Pi pin 2 (5V output) -> Pi pin 4 (5V output), FLIR Lepton VIN (5V), AHT20 VIN (5V power), BNO055 VIN (5V power)

3.3V power rail (from WittyPi)

- WittyPi 3V3 output -> MTDOT VDD (3.3V power)

Ground

- Pi pin 6 (GND) -> Pi pin 9 (GND), Pi pin 14 (GND), Pi pin 20 (GND), Pi pin 25 (GND), Pi pin 30 (GND), Pi pin 34 (GND), Pi pin 39 (GND), FLIR Lepton GND, WittyPi GND, MOSFET source, resistor lead 1, AHT20 GND, BNO055 GND, MTDOT GND

I²C Bus (shared by AHT20, BNO055, FLIR Lepton)

I²C data bus (Pi GPIO 2 / SDA)

- Pi pin 3 (I²C SDA / data) -> FLIR Lepton SDA (I²C data), AHT20 SDA (I²C data), BNO055 pin 3 (silkscreen 'SCL')

I²C clock bus (Pi GPIO 3 / SCL)

- Pi pin 5 (I²C SCL / clock) -> FLIR Lepton SCL (I²C clock), AHT20 SCL (I²C clock), BNO055 pin 2 (silkscreen 'SDA')

UART (Pi ↔ MTDOT serial communication)

Pi UART5 TX → MTDOT PA_3 (Pi sends data to MTDOT)

- Pi pin 32 (GPIO 12 / UART5 TX) -> MTDOT PA_3 (UART RX from Pi)

Pi UART5 RX ← MTDOT PA_2 (Pi receives data from MTDOT)

- Pi pin 33 (GPIO 13 / UART5 RX) -> MTDOT PA_2 (UART TX to Pi)

SPI Bus (Pi ↔ FLIR Lepton)

SPI clock to FLIR Lepton

- Pi pin 23 (GPIO 11 / SPI clock) -> FLIR Lepton SCLK (SPI clock)

SPI data: Pi → FLIR Lepton

- Pi pin 19 (GPIO 10 / SPI MOSI) -> FLIR Lepton MOSI (SPI data in)

SPI data: FLIR Lepton → Pi

- Pi pin 21 (GPIO 9 / SPI MISO) -> FLIR Lepton MISO (SPI data out)

Pi GPIO 8 / SPI chip select for FLIR Lepton

- Pi pin 24 (GPIO 8 / SPI chip select 0) -> FLIR Lepton CS (SPI chip select)

FLIR Lepton additional control signals

Pi GPIO 17 → FLIR Lepton control

- Pi pin 11 (GPIO 17) -> FLIR Lepton VSYNC / control

Pi GPIO 6 → FLIR Lepton control

- Pi pin 31 (GPIO 6) -> FLIR Lepton control

MTDOT to Pi GPIO 5 connection

Pi GPIO 5 → MTDOT PA_6

- Pi pin 29 (GPIO 5) -> MTDOT PA_6

MTDOT wake circuit (how the radio tells the Pi to power on)

The MTDOT can wake the Raspberry Pi by signalling the WittyPi. It does this through a small NMOS switching circuit:

Signal flow: **MTDOT pin 19 (PB_1) -> MOSFET gate -> MOSFET drain -> WittyPi SW pin**

When the MTDOT drives PB_1 high, the MOSFET turns on, pulling the WittyPi's SW pin to ground. The WittyPi sees this as a button press and wakes the Pi. The 10kΩ resistor on the gate keeps the MOSFET reliably off when PB_1 is idle.

MTDOT wake signal (drives MOSFET gate)

- MOSFET gate → resistor lead 2, MTDOT PB_1 (wake signal to Pi)

WittyPi wake/shutdown signal

- WittyPi SW (shutdown/wake input) → MOSFET drain

Pi GPIO pins routed but not used by any component on the hat

These Pi GPIOs are connected to the GPIO header but don't drive anything on this hat. The signal is available at the Pi but doesn't go anywhere else.

- Pi pin 8 (GPIO 14 / UART0 TX) (no destination on this hat)
- Pi pin 10 (GPIO 15 / UART0 RX) (no destination on this hat)
- Pi pin 36 (GPIO 16) (no destination on this hat)
- Pi pin 12 (GPIO 18) (no destination on this hat)
- Pi pin 35 (GPIO 19) (no destination on this hat)
- Pi pin 38 (GPIO 20) (no destination on this hat)
- Pi pin 40 (GPIO 21) (no destination on this hat)
- Pi pin 15 (GPIO 22) (no destination on this hat)
- Pi pin 16 (GPIO 23) (no destination on this hat)
- Pi pin 18 (GPIO 24) (no destination on this hat)
- Pi pin 22 (GPIO 25) (no destination on this hat)
- Pi pin 37 (GPIO 26) (no destination on this hat)
- Pi pin 13 (GPIO 27) (no destination on this hat)
- Pi pin 7 (GPIO 4) (no destination on this hat)
- Pi pin 26 (GPIO 7 / SPI chip select 1) (no destination on this hat)

Pi HAT ID EEPROM pins

Per the Raspberry Pi HAT specification, these pins must remain unconnected when no HAT ID EEPROM is fitted. They are correctly left alone here.

- Pi pin 27 (HAT ID SDA) (correctly left unconnected)
- Pi pin 28 (HAT ID SCL) (correctly left unconnected)

Pins intentionally left unconnected

These pins exist on the components but are not wired to anything else. Most are unused functions on standard breakout boards.

BNO055: BNO055 RST (reset, unused); BNO055 3V (unused); BNO055 PSI (mode, unused); BNO055 PS0 (mode, unused); BNO055 ADR (address, unused); BNO055 INT (interrupt, unused)

MTDOT: MTDOT NRESET (unused); MTDOT PA_8 (unused); MTDOT PC_9 (unused); MTDOT RFU (reserved); MTDOT PA_11 (unused); MTDOT VOUT (do not connect); MTDOT GND (currently floating!); MTDOT PA_14 (unused); MTDOT PA_13 (unused); MTDOT PB_4 (unused); MTDOT PB_3 (unused); MTDOT PA_10 (unused); MTDOT PA_9 (unused); MTDOT PB_0 (unused); MTDOT PA_5 (unused); MTDOT PA_4 (unused); MTDOT PA_1 (unused); MTDOT PC_1 (unused); MTDOT NC; MTDOT PC_13 (unused); MTDOT PA_0 (unused); MTDOT PA_7 (unused)

FLIR Lepton: FLIR Lepton pin 3 (unused); FLIR Lepton pin 4 (unused); FLIR Lepton pin 6 (unused); FLIR Lepton pin 11 (unused); FLIR Lepton pin 13 (unused); FLIR Lepton pin 14 (unused); FLIR Lepton pin 16 (unused); FLIR Lepton pin 18 (unused); FLIR Lepton pin 19 (unused); FLIR Lepton pin 20 (unused)

WittyPi: WittyPi VOUT (battery, unused); WittyPi LED (unused); WittyPi ALM (unused); WittyPi CATH (unused)

Pi: Pi pin 1 (Pi 3.3V, unused); Pi pin 17 (Pi 3.3V, unused)

Things worth checking

BNO055 SDA/SCL labels appear swapped

The schematic symbol for the BNO055 labels pin 2 as 'SDA' and pin 3 as 'SCL'. But the netlist shows BNO055 pin 2 connected to the Pi's SCL line and BNO055 pin 3 connected to the Pi's SDA line - the opposite of what the labels suggest.

Either the schematic symbol library has the labels backwards (in which case the wiring is correct), or the SDA and SCL traces really are crossed (in which case the I²C bus won't work). Worth verifying with a multimeter against the actual Adafruit BNO055 breakout silkscreen.

MTDOT pin 12 (GND) is floating

The MTDOT has two ground pins. Pin 10 is connected to GND. Pin 12 is not connected to anything. Both should be tied to ground for the LoRa radio's RF performance. Fix in v6.1.

MOSFET wake circuit gate floats at about 1.7V

When the MTDOT is idle, its pin 19 (PB_1) sits at roughly 1.7V instead of being driven cleanly high or low. That voltage is right at the IRLB8721's gate threshold, which makes the wake circuit's behaviour unpredictable. The 10kΩ pull-down resistor helps, but a higher-threshold MOSFET or an active driver would be more reliable.